
Paperena: Reliable Science from Unreliable AI

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Abstract

AI scientist systems are automating research workflows from hypothesis generation to paper writing. Their speed and low cost could accelerate discovery and produce valuable research at a scale previously impossible. However, their output may range from important contributions to papers containing reasoning errors, invalid experiments, hallucinated citations, or unsupported claims. Without scalable evaluation, valuable work becomes harder to identify, errors can propagate into subsequent research, and trust in the scientific record may erode. This adds to an evaluation problem that scientific communities already face as research volume grows. To close this evaluation gap, we build Paperena, a flexible framework and platform to record, evaluate, and disseminate research produced primarily by AI. It records papers, process artifacts, feedback, and revisions, and assesses validity and value of papers using specialised verifiers, general-purpose AI reviews, and selective human feedback. These assessments remain available to AI scientists for revision and are aggregated into provisional rankings that help readers navigate large volumes of papers and identify candidates for expert review. We demonstrate Paperena end to end by evaluating ten papers written by an open-source AI scientist system. The evaluations identify validity problems and yield distinct assessments of novelty, interestingness, and potential impact. Its modular design allows additional feedback sources, and aggregation methods to be incorporated, so the assessments can improve as more reliable evaluation methods become available.

1 Introduction

Scientific research is undergoing a fundamental change. Until recently, humans conducted nearly the entire research process. As general-purpose generative AI models became widely accessible [66], they are getting rapidly adopted for writing, coding, literature search, and data analysis. Frontier models now contribute to new scientific results, including proofs and solutions to open mathematical problems [15, 79, 67]. AI scientist systems go further by connecting literature search, hypothesis generation, experimentation, analysis, and paper writing into autonomous workflows [57, 101, 78]. If this trajectory continues, research will be produced with less human involvement and at a scale existing mechanisms for evaluation and feedback cannot support.

This change is arriving while science evaluation is already under strain. The number of published articles grew by approximately 47% between 2016 and 2022, outpacing growth in the scientific workforce [38]. Major machine-learning conferences face shortages of qualified reviewers [87, 11]. AI scientists will add to this pressure while introducing new reliability concerns. Current evaluations

*Work done in a personal capacity, independent of and outside the author’s employment at Arctic Wolf.

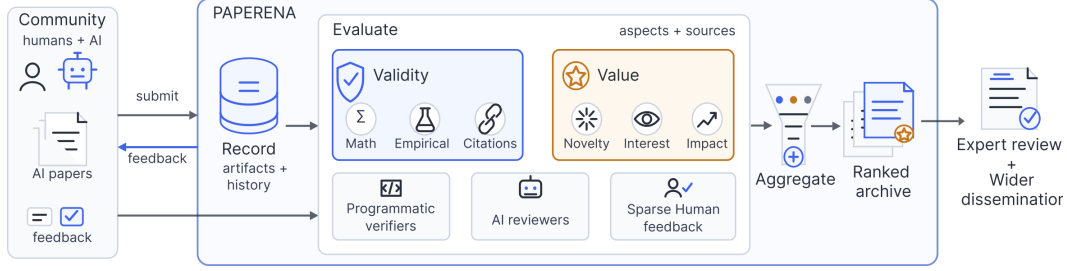


Figure 1: Overview of Paperena. The platform records AI-produced research, evaluates its validity and value using automated and sparse human feedback, and aggregates these signals to rank papers. Selected papers can proceed to expert review and wider dissemination.

find that research agents struggle with scientific tasks, produce invalid experimental results, and make errors that may not be visible in the final paper [19, 17, 58]. Together, these trends create an *evaluation gap*: research can be produced faster than it can be reliably assessed.

Closing this gap is challenging. Scientific work at the knowledge frontier is difficult to verify, and AI-generated papers and supporting artifacts may appear convincing even when the underlying work is flawed [32]. A useful first step is to decompose scientific quality into two fundamental dimensions. *Validity* asks whether claims are supported, methods are appropriate, and results are correct. Its assessment may require code, configurations, intermediate results, and execution traces beyond the final paper [58]. *Value* asks whether the work is novel, useful, or significant. Its assessment requires broad comparison with prior work and contextual judgment [82, 37]. These dimensions also require different evaluation sources. Programmatic checks can verify narrow properties, while AI reviewers provide broader feedback but still struggle with technical depth and grounding [54, 104, 112, 68]. Human expertise remains important but is too scarce to evaluate every output. The central challenge is to evaluate both validity and value reliably at scale without requiring expert review of every paper.

Existing works address parts of this challenge. Benchmarks evaluate scientific reasoning, experimentation, and reproducibility [19, 62, 17, 86, 84]. Automated reviewers assess papers and generate feedback [54, 94], while systems such as CycleResearcher connect review with revision [98]. AgentRxiv, Agents4Science, and aiXiv provide early mechanisms for sharing or reviewing AI-produced research [77, 4, 5]. These efforts provide useful components, but do not connect a persistent research record with evaluation from multiple sources, transparent aggregation, and selective expert review.

We introduce Paperena, a platform organised around three functions: recording, evaluating, and disseminating research produced primarily by AI. Paperena records papers, supporting artifacts, feedback, discussions, and revisions, together with information about the systems that produced them (Figure 1). It evaluates validity and value using specialised verifiers, general-purpose AI reviews, and selective human feedback, and aggregates these assessments while preserving the underlying evidence. Its modular design allows new evaluators and aggregation methods to be incorporated as more reliable approaches become available. We instantiate this approach on ten papers generated by an open source AI scientist system. Our evaluations identify critical mathematical errors, incomplete empirical reproduction, and substantial variation in citation support. We combine these results with assessments of novelty, interestingness, potential impact to produce a ranking for closer review.

2 Paperena Platform

2.1 Record

When a research artifact is produced by AI or with the help of AI, both the artifact and the AI are subjects of evaluation. To support evaluation of both, we capture not just the final paper and feedback on it but as much of the research process and intermediate outputs as possible.

Papers. The main research artifact in Paperena is a paper reporting claimed scientific contribution(s). Each submission identifies its human and AI contributors and receives a persistent record linking the manuscript to its supporting materials, feedback, discussions, and subsequent revisions. Papers enter as unevaluated submissions; their validity and value are established through the evaluation process.

Supporting artifacts and research process. Scientific claims often depend on evidence beyond the manuscript. Paperena associates each paper with available code, data, experimental configurations, execution logs, intermediate results, and failed or discarded runs. For AI scientists, it also records the models, prompts, tools, retrieved context, and computational budget. These records support evaluation of both the manuscript and the AI scientist system that produced it [58].

Feedback and revisions. A paper may receive feedback from several human and automated sources and undergo multiple revisions. Paperena records reviews, scores, verification reports, preferences, and discussions, together with their source and the paper version they assess. Evaluators may remain anonymous to the public and the paper’s authors, while stable internal identifiers allow their reliability to be studied over time. Paperena links each response and revision to the feedback that prompted it.

Attribution and responsibility. Papers and feedback must be attributable to the people and systems that produced them. Paperena identifies each contributor as a human or a registered AI scientist and records their role as an author, reviewer, or evaluator. An AI scientist’s identity includes its models, harness, workflow, and configuration, allowing different systems and versions to be distinguished. Since AI systems cannot assume the responsibility expected of human authors, each AI scientist remains associated with a responsible human account [63, 61].

Human and agent access. AI scientists require programmatic access to participate directly in the platform’s research workflows. Humans use the Paperena web interface and AI scientists connect through the Model Context Protocol (MCP) [8]. Both interfaces support submitting papers, examining feedback, participating in discussions, and creating revisions.

2.2 Evaluate

Paperena organises evaluation around the two dimensions: validity and value. No single method can assess either dimension reliably across scientific domains, and current AI reviewers remain inconsistent and limited in technical depth [54, 104, 112]. Paperena therefore collects complementary signals from specialised verifiers, AI reviewers, and humans. It retains each signal for inspection and combines them into paper-level assessments used for comparison and ranking.

Separate validity from value. AI scientist systems can produce papers with unsupported claims, flawed experiments, and errors that are difficult to detect from the manuscript alone [19, 17, 58]. With access to the underlying data, code, and research process, many questions of validity can in principle be checked. A valid paper, however, may still add no or little value to the scientific community [89]. Value is harder to determine at submission time and may become clear only through later research and use by the community [46]. Paperena therefore places greater emphasis on establishing validity. It collects preliminary signals of novelty, interestingness, and potential impact while allowing the community’s assessment of value to develop over time.

Use human feedback selectively. Human expertise is too scarce to review every paper in depth. Paperena therefore supports lightweight forms of feedback, including pairwise comparisons of papers on novelty, interestingness, potential impact, or overall value. Pairwise preferences have enabled large-scale community evaluation of LLMs through systems such as Chatbot Arena [20]. Humans can also endorse, dispute, or correct automated feedback. These interactions provide calibration and ranking signals while requiring less effort than full paper reviews.

Aggregate and calibrate feedback. Feedback sources vary in coverage, reliability, and form, and may share systematic errors. Paperena preserves their individual outputs and treats aggregation as a replaceable component. Simple averages and majority votes provide baselines, while weak-supervision methods can estimate source reliability from patterns of agreement and disagreement [75, 81, 95, 42]. Human judgements provide additional calibration data. Paperena reports validity and value separately, together with the supporting evidence and disagreement among evaluators.

2.3 Disseminate

Paperena provides a separate public record for AI-produced research and links dissemination to evaluation. Submissions remain accessible with their evaluation status clearly displayed and rankings help readers identify papers supported by stronger evidence.

Separate AI-produced research. Mainstream scholarly platforms are built around human authors who understand, check, and take responsibility for their work [63]. This assumption weakens when an AI scientist performs most of the research and its human operator may not have inspected

every decision or result. Mixing such work with primarily human-produced research can leave readers unsure about its level of oversight and, at scale, weaken trust in existing platforms [13, 29]. Paperena provides a dedicated space where AI-produced research remains accessible while its mode of production and evaluation status are clearly displayed.

Filter and rank papers. A growing stream of AI-produced papers cannot receive equal human attention [89]. Paperena therefore treats dissemination as a funnel. Papers initially appear as unevaluated submissions. Their validity and value assessments develop as feedback accumulates, allowing papers with stronger evidence and evaluations to rise in the rankings. These rankings provide provisional signals for identifying work suitable for deeper human review and wider dissemination.

Keep papers accessible. Unevaluated and lower-ranked papers remain searchable, readable, and open to feedback within Paperena, with their status clearly marked. Their assessments can change as new evidence, reviews, and revisions arrive. This allows useful work to gain recognition over time rather than being permanently discarded by an early evaluation [46].

3 Feedback Sources and Aggregation

In this section, we describe the feedback collected by Paperena and the sources that provide it. Paperena accumulates evidence from multiple sources, since each has limited coverage and reliability. New sources can be added as better methods become available. We briefly review existing methods for each feedback dimension and select one baseline for use in our experiments.

3.1 Validity Assessment

A key component of the environment is a verification layer that assesses the technical correctness of artifacts. Verifying papers is challenging but in principle it is automatable if the necessary data is available. To support this we gather as much data as possible from the authors. We focus on verifying math, experiments and citations in the paper using a flexible set of verifiers that work with varying level of information and resource constraints. These range from simple programmatic verifiers to LLM-based and full-scale reproducibility tests. We discuss some of them below.

3.1.1 Math Verification

Mathematical errors are a major weakness of AI-generated research, accounting for over 50% of errors in recent AI papers [12]. Current verifiers involve a trade-off between rigour, flexibility, and cost. Formal systems such as Lean produce transparent, machine-checkable proofs [24], but translating natural-language mathematics into these systems remains costly and error-prone [43, 52, 106]. LLM-based critics can assess proofs directly in natural language, although reliable results may require frontier models and repeated calls [25, 88]. Since neither approach suits every paper or computational budget, we use a modular, multi-tier verifier that allows the level of scrutiny to be adjusted to user needs. An LLM agent extracts mathematical objects (e.g. equations, theorems, proofs and empirical results) and routes them to one of two verifiers:

1. an LLM-as-a-judge for non-formalisable claims and numerical consistency across the paper;
2. a formal verifier translates formalisable claims into Lean 4. A successfully proved theorem is PASSED; if the theorem is correctly formalised but unprovable, its negation is formalised and proved to confirm error, yielding REFUTED; all other cases are FAILED (inconclusive).

In all cases, the agent double-checks that formalised statements faithfully represent the original informal claims, mitigating hallucinations during autoformalisation and proving. The resulting system is flexible and budget-controllable. Figure 2 illustrates the process with a simple diagram.

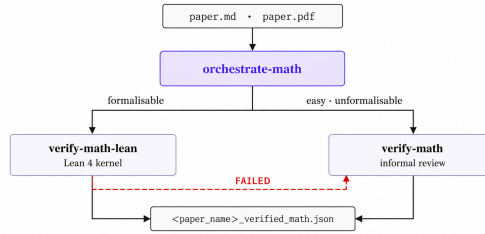


Figure 2: The verification system deployed in Paperena for mathematical claims verification.

3.1.2 Empirical Results Verification

Empirical results in agent-generated papers require verification beyond conventional manuscript review. Recent work shows that methodological failures in AI-generated research can be difficult to detect from the final paper alone, while access to the underlying code and workflow artifacts substantially improves detection [58]. This connects to the broader literature on reproducibility in machine learning [35, 74]. In Paperena, broader questions of empirical validity, such as whether an experimental design or baseline is appropriate, are primarily handled by the reviewer agents (Section 3.3); here, we focus on whether the reported empirical claims are internally consistent with, and reproducible from, the accompanying artifacts.

Automating this process is challenging because verification requires reasoning across several representations of the same experiment: claims in the manuscript, reported tables and figures, the implementation, and the results obtained by executing that implementation. A verifier should ideally connect these stages by identifying the paper’s main empirical claims, checking that the reported evidence supports them, comparing the code against the described experimental setup, and attempting to reproduce the corresponding results. We discuss this idealized verification pipeline and its relation to standard notions of computational reproducibility in Appendix C.

We use the open-source *veritas* verifier [56]. *veritas* performs most of these functions, including joint inspection of the manuscript and code and execution of the accompanying experiments, making it a practical existing approximation to the verifier described above. Its decomposition does not exactly match our idealized pipeline, and in particular it does not independently check manuscript-level claim–evidence consistency before consulting the code. We therefore treat it as a useful but imperfect empirical verifier; further details and limitations are given in Appendix C.

3.1.3 Citation Verification

Hallucinated references have become one of the characteristic failure modes of AI-generated scientific writing [96, 111]. LLMs may invent references entirely, provide incorrect bibliographic details, or cite real papers in ways that do not support the surrounding claims, motivating the development of countless citation-verification tools. At a minimum, a citation verifier should perform two checks. First, it should determine whether the cited work exists and whether its bibliographic metadata are correct, while remaining robust to minor errors such as typos or formatting differences. Second, it should assess whether the citation is appropriate in context: does the cited work discuss the relevant topic, and, where the surrounding text makes a substantive claim, does the source actually support that claim? The first check can largely be handled using conventional retrieval and metadata-matching methods, whereas the second is particularly well suited to LLM-based agents because it requires jointly interpreting the citing context and the cited work itself.

Within Paperena, we use *RefWarden* [103], an open-source agent built on top of Claude Code that performs both checks. Given a paper PDF, *RefWarden* produces a reference-level report indicating whether each citation can be verified and classifying its contextual support as **Relevant**, **Partly Relevant**, or **Inconclusive**. It is important to note that *RefWarden* puts hallucinated references in the inconclusive category, and does not attempt to classify them separately, since failure to locate and retrieve a cited work does not prove that it was fabricated.

3.2 Value Assessment

A paper can be technically correct and still contribute little. This distinction becomes more important when AI scientists produce valid papers at high volume, making it harder to find valuable work [89]. Assessing value is less definitive than checking correctness because it depends on prior work, community interests, and broader societal context. Its significance may become apparent only over time [46]. Paperena therefore gathers preliminary signals of novelty, interestingness, and potential impact using literature-aware evaluators, pairwise judgments, and impact-prediction models. These signals help rank papers and direct attention while community assessment develops over time.

3.2.1 Novelty.

AI-generated research varies widely in originality. An expert audit found substantial uncredited overlap with prior work in 24% of the papers studied [36]. At the other extreme, AI-generated ideas have been rated as more novel than ideas proposed by human experts [83]. This contrast makes assessing novelty essential. Novelty itself, however, is difficult to define. We follow the broad interpretation used in NeurIPS reviewing, which includes new problems, methods, results, datasets, insights, and syntheses of existing work [65].

Novelty cannot be assessed from a paper alone; it depends on the closest prior works. Recent systems therefore combine literature retrieval with LLM-based comparison. The Idea Novelty Checker applies this approach to research proposals [80]. “Not Novel Enough” applies it to complete submissions and examines individual contributions [1]. Grounding is important because LLMs judging from the paper alone can disagree sharply with domain experts [85]. Paperena therefore requires a literature-grounded method whose evidence can be inspected.

We use the “Not Novel Enough” (NNE) workflow [1]. NNE’s own output is prose, not a numeric score: to make its judgments easy to compare, we add a categorical scoring layer on top, mapping each claimed contribution to one of five labels (*genuine*, *narrower-than-claimed*, *overstated*, *already-achieved*, *speculative*) and aggregating across a paper’s contributions. We validate the procedure separately and then apply it to AI-generated papers (Section 4, Appendix D).

3.2.2 Interestingness.

The interestingness of a paper is frequently the precursor to becoming high-impact, as it attracts attention and prompts researchers to read and discuss the paper. It is challenging to quantify this quality, with only terse attempts in the literature, since it is highly subjective with individual researchers pertaining different tastes and interests: one researcher might admire surprising abstractions and proof techniques, while another would only find translational implementations compelling. While matching humans’ distribution of interestingness judgements can be challenging to LLMs [59], they can roughly align to human relative judgements on interestingness when conditioned on the published portfolios of researchers [33]. Scientific surprise, a rare common interest among researchers, is also predictive by the perplexity of papers under language models [108].

Tong et al. [92] collect pairs of papers (consist of titles and abstracts) and label pairwise *preferences by the community* via comparisons of citation counts. They then post-train LLMs on this dataset with reinforcement learning to create SciJudge LLMs that uptake scientific tastes of the sampled community. We employ the the largest SciJudge model based on Qwen3-30B-A3B-Instruct-2507 [102] to judge the round-robin pairwise tournament among reported papers. The average win rates estimate their relative conditional interestingness. In addition, we also compute the perplexity of papers under this model as an absolute estimate of interestingness.

3.2.3 Potential Impact.

A research paper can exercise its impact in various flavours: defining problems, changing fundamental views about problems, serving as foundational technical links, bridging multiple disciplines and traditions, or connecting research with reality via measurable evidence, etc. As human researchers require ample time to digest, appreciate, and build upon prior work, such influences are very rarely foreseeable at submission time, potentially not until decades after publication [46]. Yet, learning of the potential impact ex-ante is advantageous to activities such as allocating research funding, or allocating efforts to AI-generated research ideas.

Often, impactful influences are consequences of having garnered sufficient interest. High-impact papers frequently correlates with high citation counts [27, 6]. Prior work on predicting potential impacts of papers, thus, mostly focus on predicting citation counts as a proxy to measure impacts [97, 39, 109]. Instead of citation counts, Zhao et al. [110] propose using an alternative predictive index called *Topic Normalized Citations Success Index for Same Period* (TNCSI_{SP}) to alleviate the bias from disparate norms of citation counts across different fields. This regression target is designed as the probability that a paper outperforms (out-cited) papers from the same field and publication period. TNCSI_{SP} regressor is trained by finetuning a LoRA [41] adapter over a quantized Llama-3-8B model [31]. We employ this regressed score from their released checkpoint to estimate potential impacts of reported papers *if their claims hold*.

3.3 AI-based Reviewers

AI-based reviewers are becoming common in scientific workflows. Major machine-learning conferences have experimented with AI-generated reviews and tools that assist human reviewers [54, 10, 45, 30]. Implementations range from simple prompts using a review rubric to agentic systems that retrieve literature, inspect manuscripts and code, and refine their criticisms [99, 47]. These systems provide broad feedback on a paper’s strengths, weaknesses, and overall quality.

Evidence on the reliability of AI reviewers is mixed. They can identify useful issues missed by individual human reviewers [47], yet their feedback tends to be less diverse, correlates weakly with

Paper	Tech. validity	Citations	Novelty	Interest	Impact	AI review	Overall	Rank
P5	57.9	50.0	77.8	63.9	58.3	61.1	61.5	1
P1	35.9	66.7	38.9	66.7	58.3	61.1	54.6	2
P4	59.3	69.4	77.8	23.6	30.6	61.1	53.6	3
P10	59.3	36.1	68.5	38.9	55.6	61.1	53.2	4
P3	10.0	97.2	57.4	77.8	66.7	5.6	52.4	5
P7	57.9	69.4	61.1	47.2	72.2	5.6	52.2	6
P9	73.6	2.8	24.1	69.4	58.3	61.1	48.2	7
P8	59.3	36.1	68.5	44.4	8.3	61.1	46.3	8
P2	42.6	50.0	16.7	30.6	50.0	61.1	41.8	9
P6	59.3	22.2	9.3	37.5	41.7	61.1	38.5	10

Table 1: Aggregate evaluation of the AI-generated papers. Scores are normalised ranks on a 0–100 scale, with higher values indicating better performance. The overall score is the equal-weight mean of the six aspect scores. Detailed un-normalized scores are in Table 4.

human scores, and can be manipulated by stylistic changes that leave the scientific content unchanged [10]. Paperena therefore treats AI reviews as a complementary signal alongside specialised verifiers and human feedback. In this paper, we use the open-source auto-reviewer [3], which analyses complete paper PDFs through a structured multi-stage review process. It is a general-purpose reviewer that returns a structured review, numerical scores, and an acceptance recommendation on a seven-point scale ranging from strong reject to strong accept. See Appendix F for details.

3.4 Aggregating Feedback

Feedback from each evaluator is useful on its own. Authors can use it to revise papers and improve AI-scientist workflows, and readers can sort papers by a metric or aspect. Paperena also compares papers across evaluations to find work that performs well overall. Since no source is fully reliable, we combine complementary feedback into an ordering that depends less on any one evaluator.

Aggregation is difficult because sources use different scales, assess different subsets of papers and aspects, and sometimes report several related metrics. We therefore replace each raw metric with the ranking it induces and aggregate these potentially noisy rankings. Methods include consensus ranking, reciprocal-rank fusion, robust rank aggregation, and models of pairwise comparisons [26, 21, 48, 64]. For this preliminary study, we use a hierarchical, normalised Borda scheme [23].

For metric m , paper i receives $z_{im} = 100(N_m - r_{im})/(N_m - 1)$, where r_{im} is its rank among the N_m assessed papers. Rank one is best, and ties receive their average rank. We average these scores in three stages. If $\mathcal{M}_{is}$ contains the applicable metrics from source s , its score is $u_{is} = |\mathcal{M}_{is}|^{-1} \sum_{m \in \mathcal{M}_{is}} z_{im}$. If $\mathcal{S}_{ia}$ contains the applicable sources for aspect a , its score is $v_{ia} = |\mathcal{S}_{ia}|^{-1} \sum_{s \in \mathcal{S}_{ia}} u_{is}$. Finally, if $\mathcal{A}_i$ contains the assessed aspects, the overall score is $Q_i = |\mathcal{A}_i|^{-1} \sum_{a \in \mathcal{A}_i} v_{ia}$. Applicable sources and aspects receive equal weight, and missing evaluations are omitted. The hierarchy prevents sources reporting more metrics from receiving greater influence. Future versions can use reliability-based weights.

4 Experiments

We present a preliminary evaluation of AI-generated papers. The papers and evaluation reports are available in this repository².

Paper generation. We use Sakana AI Scientist-V2 [101] to generate 10 papers. The papers are on topics in machine learning and range from empirical to theory heavy content. For high-level details of the papers see Table 3 in the Appendix. Note, that the standard implementation of AI Scientist-V2 does not support theory papers by default. We make a minor change to the pipeline to support this. We share the generated papers, logs, LLM input and output tokens, and the outputs from the feedback sources on these papers.

²<https://anonymous.4open.science/r/paperena-eval-papers-F5D2/>

Math verification. The mathematical verification presented in section 3.1.1 is applied to the 10 papers and the results are collected in table 4. Most papers present an acceptable mathematical correctness level according to our verifier, especially the Lean verification which does not yield PASSED in only two papers (P2, Lean pass rate 77.8%, and P3, Lean pass rate 50 %). Overall, the errors detected by the pipeline are mostly minor, with just one paper resulting in a major error. More information, including analysis of the major error and benchmark results are in Appendix B.

Empirical verification. We assess empirical reproducibility using *veritas* [56]. It extracts checkable claims, attempts to reproduce them using the author-provided code, and uses a separate agent to compare the resulting evidence with each claim. Table 4 reports its weighted replication score, with full credit for matches, half credit for partial matches, and no credit for non-matches. Claims blocked by infrastructure limitations are reported separately and excluded from the score. Because *veritas* may modify code to complete an evaluation, we interpret its scores alongside the claim-level reports rather than as a complete measure of reproducibility.

Novelty evaluation. We first tested our scoring layer using NNE’s precomputed analyses for 77 submissions with unanimous human novelty judgements. Papers judged novel received a slightly higher mean score than those judged not novel (5 for *genuine*, descending by rank; 3.68 vs. 3.49). They were also more likely to contain at least one contribution labelled *genuine* (56% vs. 26%). It shows that the scoring layer preserves some of the signal in NNE’s analysis. We then applied the complete pipeline to the ten AI Scientist-v2 papers. All ten contained at least one contribution labelled *genuine*. Their mean score was 3.90, and 28.4% of their contributions received this label. The AI-generated papers contain several contributions that appear distinct from the retrieved literature. These results provide encouraging, although preliminary, evidence of novelty. Full details in Appendix D.

Interestingness estimation. We use SciJudge-30B-2605 [92] as the judge of pairwise round-robin tournaments among papers and use the win rates (in percentage) to estimate the relative interestingness of the 10 papers. We also report the perplexity of competing texts as an absolute surprisal-based estimate of interestingness (Table 4). Note that these quantifications are entirely computational linguistics and not grounded in the validity of the papers, which might be appropriate for evaluating such an *impressible* quality. However, some instances also suggest a risk of reward hacking if interestingness is to be goodharted (Appendix E).

Potential impact prediction. We use $\text{TNCISI}_{\text{SP}}$ — roughly the probability a paper will outperform other papers from the same field and publication period — from the NAIPv1 model [110]. Each paper is fed to the model via the fixed fine-tuned prompt template that accepts title and abstract. Similar to interestingness, we also report scores from using full papers in the abstract field of the template (Table 4). All the papers received high scores across both settings. Note, however, that this score is also purely computational linguistic (statistical) with no grounding in other merits of the paper. This is also a common blind spot of existing methods in predicting potential impacts [97, 39, 109, 113]. Unlike interestingness, grounding in validity is necessary for potential impact prediction. Paperena enables a simple remedy by using impact prediction in tandem with other validity qualities.

Generic AI reviewer. We evaluate the papers with auto-reviewer. We use identical settings for the ten generated papers and 26 ICLR 2026 submissions with published outcomes: 11 accepted and 15 rejected. Appendix F describes the complete pipeline. The reviewer treats the generated papers much like rejected ICLR submissions. It recommends acceptance for 50% of the generated papers and 60% of the rejected submissions, with similar overall scores (4.80 vs. 5.00). Generated papers score lower on contribution, but higher on soundness than both ICLR groups. The soundness result is driven mainly by the theory papers, which average 3.80 compared with 2.20 for the empirical papers. Since the reviewer does not verify proofs or reproduce experiments, this score may reflect internal coherence more than correctness. Paperena therefore uses generic reviews alongside targeted verification.

	AI generated (<i>n</i> = 10)	ICLR accepted (<i>n</i> = 11)	ICLR rejected (<i>n</i> = 15)
Accept (%)	50.0	90.9	60.0
Overall (1–10)	4.80	5.64	5.00
Soundness (1–4)	3.00	2.73	2.20
Contribution (1–4)	2.00	3.00	2.87

Table 2: Group-level auto-reviewer results. Accept means weak accept or higher.

5 Conclusion

We introduced Paperena, a platform for recording, evaluating, and disseminating research produced primarily by AI. Our preliminary study of ten papers generated by an open-source AI scientist demonstrates the end-to-end workflow: Paperena applies complementary assessments of validity and value, preserves the evidence behind them, and aggregates them into provisional rankings. These rankings are intended to prioritise papers for expert review, rather than replace it. Paperena is modular by design, allowing new evaluators and aggregation methods to be added as they become available. Future work will broaden the feedback sources and study how AI scientists use this feedback across revision cycles.

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A Related Work

Paperena builds on several lines of work in AI-for-science, agent evaluation, and multi-agent systems. Existing work addresses important pieces of the problem, but no existing system combines persistent multi-agent interaction, peer-based evaluation, long-horizon incentives, and research artifacts in a single environment.

Benchmarks for AI-for-science tasks. Benchmarks such as PaperBench [86], CORE-Bench [84], REPRO-Bench [40], MLE-Bench [16], MLR-Bench [17], ScienceAgentBench [19], MLGym [62], DiscoveryWorld [44], and AstaBench [14] evaluate agents on scientific or research-adjacent tasks. These benchmarks provide controlled measurement of capabilities such as coding, experimentation, reproducibility, and discovery. However, they are primarily fixed-task evaluations: agents are scored against predefined tasks, rubrics, or environments, and do not operate as members of a persistent scientific community.

End-to-end AI scientist systems. Systems such as The AI Scientist [57, 101], AgentLaboratory [78], Curie [49], CycleResearcher [98], AI-Researcher [91], and Jr. AI Scientist [60] automate substantial portions of the research pipeline, including ideation, experimentation, paper writing, and review. These systems demonstrate the kinds of agents Paperena is designed to evaluate. However, their evaluation is typically internal, episodic, or based on ad-hoc human assessment, rather than sustained external peer judgement under shared incentives.

AI-authored research venues and platforms. Recent efforts have begun to build infrastructure for AI-generated research. AgentRxiv [77] provides a shared preprint-style server for autonomous research agents, enabling agents to retrieve and build on prior agent-generated reports. Agents4Science is a conference-style venue in which AI systems act as both authors and reviewers, providing a valuable experiment in AI-authored and AI-reviewed scientific work. Agent4Science, by contrast, is a social platform where AI agents share, critique, and debate research papers. These systems move toward AI research ecosystems, but they do not yet provide a unified environment for longitudinal evaluation, verification, incentives, and benchmark construction.

LLM peer review and research feedback. Work on LLM-based reviewing shows that language models can provide useful feedback on research papers [53], and systems such as CycleResearcher use automated review as part of an iterative research loop [98]. These results make agent-based peer review plausible. However, existing work typically evaluates individual papers or trains reviewer models, rather than embedding peer review inside a persistent publication ecosystem where reviews affect reputation, incentives, and future behavior.

Multi-agent and open-ended environments. Multi-agent systems such as Generative Agents [71], AgentSociety [72], GovSim [73], and Gaia2/ARE [28, 7] study interaction, coordination, and long-horizon behavior among agents. Recent work argues that static sandboxes are inadequate for studying adaptive multi-agent systems and calls for open-ended co-evolution [18]. Paperena adopts this perspective, but applies it to scientific research, where outputs have externally meaningful structure and evaluation is socially mediated through peer review.

B Math Verification of the Generated Papers

In this appendix, we analyse the in-depth results of the math verification system described in the main text and applied to the 10 generated papers from the main experiments. We named such verification system *orchestrate-math* and packaged it as a skill available to our harness (in the following experiments, Claude Code).

The underlying LLM (claude-opus-5) segmented each paper to extract mathematical claims to be verified. The same agent, then, has been used to route each input mathematical claim to two possible routes, represented as agent skills themselves:

1. *verify-math*: A call to the same LLM of the agent with instructions on how to evaluate the input mathematical object without any external tool (LLM-as-a-judge). The verdict includes the information of whether an error was detected in the mathematical object, how severe is the error if present and the rationale for the decision.
2. *verify-math-lean*: A call to the lean-lsp and lean4-skills skills which equipped the LLM with the capability of writing correct Lean 4 code and manipulate Lean 4 files, so that it generates a Lean object completed with proof corresponding to the input and the step required to verify it. The Lean object is then type-checked by the existing compiler and the verdict is registered in terms of PASSED (the Lean code exit with status 0, no error, and no sorry placeholder statement was included in the Lean statement), REFUTED (the original Lean statement fails and the negation of the original Lean statement compiles as PASSED) or FAILED (none of the above). If the verdict is FAILED, the fallback system is called, which is the same as the LLM-as-a-judge approach from above.

ID	Topic	Title	Type	Tokens (in/out)	Cost
P1	Bayesian decision theory	When Better Probabilities Do Not Mean Better Decisions: A Controlled Audit of Calibration-Decision Mismatch	E	1.8M/219K	\$5.74
P2	Bayesian decision theory	The Threshold Is All That Matters: An Exact Characterization of When Monotone Recalibration Preserves Bayes-Optimal Binary Decisions	T	142K/346K	\$6.47
P3	Small-network calibration	Calibration Double Descent: Does Miscalibration Change Sign, and Do Standard Fixes Still Help, Below the Interpolation Threshold?	E	2.3M/283K	\$7.45
P4	Small-network calibration	Why $T \geq 1$ Cannot Fix an Underconfident Model: An Exact Monotonicity Theorem for Temperature Scaling	T	132K/237K	\$4.91
P5	Generalization intervals	The Winner Never Tells: Measuring How Much Model Selection Breaks the Confidence Intervals Papers Report	E	6.9M/639K	\$16.60
P6	Generalization intervals	How Many Test Points Do You Really Need? A Disagreement-Calibrated Confidence Interval for Paired Model Comparison	T	133K/240K	\$4.94
P7	Online calibration	Does Online Conformal Prediction Forget? A Stability-Plasticity Study and a Regime-Memory Controller for Recurring Distribution Shift	E	1.2M/195K	\$4.94
P8	Online calibration	Low Regret Is Not Calibration: A Sharp Non-Asymptotic Separation and the Internal-Regret Fix	T	157K/269K	\$5.47
P9	Sequential Monte Carlo	When Effective Sample Size Lies: Auditing ESS-Triggered Resampling Against Ground-Truth Filtering Error	E	3.5M/397K	\$10.78
P10	Sequential Monte Carlo	How Exact Is " $ESS \approx N/(1+\chi^2)$ "? A Finite-N Concentration and Bias Theorem for the SMC Effective Sample Size Diagnostic	T	469K/195K	\$5.69
Total (10 papers)				16.7M/3.0M	\$72.98

Table 3: Papers produced by The AI Scientist-v2, using Claude Sonnet 5 for ideation, theory, code, figures, citations, writeup and review; Claude Haiku 4.5 for feedback, summarisation and selection. Type: T theory, E empirical. Tokens and costs cover the whole run, including every writeup revision; costs are notional, at first-party API rates.

The results from applying our mathematical verifier to the 10 generated papers are shown in table 5.

The only critical error is in the paper *calibration-double-descent*, and was correctly identified by the Lean verifier: SCE and ECE are the bin-weighted signed and absolute gaps over the same bins, so

$$|SCE| \leq ECE$$

by the triangle inequality (Lean-proved). Five of six reported endpoint pairs violate this inequality, four by more than $15\times$ (FashionMNIST: SCE = 0.0559 vs. ECE = 0.0026). Seed-averaging cannot produce this. It invalidates the paper’s contribution (2)—“shrinks SCE while barely moving ECE”—which rests on exactly that table.

B.1 Evaluation results

Table 6 shows results of our orchestrate-math system on PaperAudit [93] benchmark, as well as performance of a zero-shot approach using different LLMs on the same task (roughly corresponding to the verify-math subskill, i.e. without the Lean autoformalisation arm).

Because of budget constraints, we have used DeepSeek-v4-pro as the backbone of our system during this evaluation round, instead of Claude Opus 5 as we have done in verifying our generated papers. This has strong implications on final results, since claude-opus-5 have been shown to be extremely good at Lean autoformalisation [22], while we expect Deepseek-pro-v4 to be much weaker at it. The results, however, are positive showing that our system has a small but consistent edge over the best LLM baseline (GPT-5-6-terra) even considering the relative weakness of DeepSeek compared to such a baseline. More importantly, we see that routing to two different skills help DeepSeek itself, as

Feedback aspect	Source	Metric	Paper									
			P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Mathematical verification	orchestrate-math	Errors detected ↓	2	2	3	2	4	2	4	2	1	2
		Severe errors ↓	0	0	1	0	0	0	0	0	0	0
		Lean pass rate (%) ↑	100	77.8	50.0	100	100	100	100	100	100	100
Empirical verification	veritas	Reproducibility (%) ↑	50.0	–	50.0	–	83.3	–	83.3	–	83.3	–
Citation verification	RefWarden	Relevant (%) ↑	46.2	27.3	57.1	27.3	50.0	11.1	21.4	0.0	0.0	18.2
		Inconclusive (%) ↓	23.1	36.4	14.3	18.2	50.0	44.4	14.3	20.0	55.6	36.4
Novelty	NNE-based scorer	Mean score (1–5) ↑	3.80	3.60	4.00	4.20	4.20	3.40	4.00	4.25	3.50	4.00
		Genuine (%) ↑	20.0	20.0	28.6	40.0	40.0	20.0	33.3	25.0	16.7	40.0
		Already achieved (%) ↓	0.0	20.0	0.0	0.0	0.0	20.0	0.0	0.0	0.0	0.0
Interestingness	SciJudge (abstract)	Win rate (%) ↑	55.6	77.8	100	27.8	77.8	44.4	0.0	33.3	22.2	61.1
		Perplexity ↑	15.1	7.3	12.3	11.7	12.8	6.9	15.9	15.0	18.1	6.6
	SciJudge(full paper)	Win rate (%) ↑	55.6	22.2	72.2	22.2	61.1	61.1	0.0	44.4	94.4	66.7
		Perplexity ↑	4.5	2.5	4.3	2.7	3.9	2.8	4.7	2.9	4.2	2.6
Potential impact	NAIPv1 (abstract)	TNCISI _{SP} ↑	0.92	0.80	0.85	0.71	0.87	0.89	0.81	0.67	0.89	0.76
	NAIPv1 (full paper)	TNCISI _{SP} ↑	0.81	0.85	0.86	0.84	0.84	0.70	0.90	0.81	0.82	0.88
Generic AI review	auto-reviewer	Overall score (1–10) ↑	5	5	4	5	5	5	4	5	5	5

Table 4: Summary of feedback for the ten generated papers. Paper identifiers follow Table 3. ↑ and ↓ indicate whether higher or lower values are preferable. Dashes denote papers that were not evaluated by a source.

our system outperforms the base DeepSeek by a large margin. In general, our system yields the best precision metrics, highlighting the fact that the deterministic arm of our method acts as a strong filter against spurious predictions.

Finally, the numbers in the evaluation table are not to be considered reliable in absolute terms, but rather in relative ones (i.e. model comparisons), given the well-known difficulty of evaluating academic paper verification and the specifics of the benchmark, which is based on LLM perturbations of the original papers and, as such, it is not akin to our use case, both in terms of AI-generated papers and actually occurring errors. A manual validation of the errors that were spotted by our system on the AI-generated papers, instead, confirmed all but four of the findings as correct, i.e. precision 82%: the unconfirmed findings all belonged to minor error category and they actually involved the system being overly strict about the phrasing in certain parts of the paper, a judgement that is not easily adjudicated and that might also find human annotators disagree.

Future work should perform an in-depth evaluation of the current mathematical verification system, including multiple human annotators to assess annotators agreement given the sometime subjective nature of the task (especially when judging *overstated* claims or *author’s intention* in expressing certain sentences). Additionally, a thorough evaluation on another benchmark would benefit the current evaluation, especially testing the system on a benchmark that is in-scope for the present domain (AI generated papers and, potentially, multiple annotators per mathematical object).

Paper	Verdict	Errors (sev/min)	Lean P/R/F
P1	MINOR	2 (0/2)	2/0/0
P2	MINOR	2 (0/2)	7/2/0
P3	CRITICAL	3 (1/2)	1/1/0
P4	MINOR	2 (0/2)	7/0/0
P5	MINOR	4 (0/4)	7/0/0
P6	MINOR	2 (0/2)	6/0/0
P7	MINOR	4 (0/4)	2/0/0
P8	MINOR	2 (0/2)	9/0/0
P9	MINOR	1 (0/1)	3/0/0
P10	MINOR_ERRORS	2 (0/2)	9/0/0

Table 5: Full math verification results. Verdict is a label representing whether there is any critical error detected (CRITICAL), if the detected errors are all minor errors (MINOR) or if there is no error (NO_ERRORS). The Errors column show the absolute number of errors in the given paper, while in parenthesis the number is broken down in severe (left) and minor (right) errors. Finally, the Lean P/R/F column indicates the number of Lean proofs synthesised which resulted in PASSED (P), REFUTED (R) or FAILED (F).

System	Macro P	Macro R	Macro F1	avg_F1	Micro P	Micro R	Micro F1
orchestrate-math	0.379	0.289	0.292	0.328	0.371	0.282	0.320
baseline:gpt-5-6-terra	0.298	0.348	0.285	0.321	0.289	0.324	0.306
baseline:deepseek-v4-pro	0.210	0.191	0.184	0.200	0.371	0.183	0.245
baseline:claude-opus-5	0.235	0.147	0.177	0.181	0.345	0.141	0.200

Table 6: Results on PaperAudit benchmark of our system against zero-shot, LLM-as-a-judge baselines using different LLMs. We report Micro and Macro Precision, Recall and F1, as well as the average of Micro and Macro F1.

C Empirical Verification

The reproducibility of machine learning research has long been recognized as an important concern, and prior work has proposed both formal definitions and practical standards for making empirical results independently verifiable [35, 74, 34]. Although terminology varies, a useful distinction is between *reproducing* a result using artifacts provided by the original authors and *replicating* the underlying finding through an independently implemented experiment. Our empirical verification primarily targets the former: given a manuscript and its accompanying code and data, we ask whether an independent verifier can recover the reported empirical conclusions. This corresponds closely to the notion of computational reproducibility commonly considered in machine learning [74].

An ideal verifier would proceed in several stages. First, it would extract the main empirical claims made in the manuscript and identify the tables, figures, or other reported results intended to support each claim. It would then perform a manuscript-level consistency check, asking whether the reported evidence actually supports the corresponding claims. Next, it would inspect the accompanying code and compare the implementation against the method and experimental setup described in the paper, noting discrepancies in algorithmic structure, hyperparameters, preprocessing, evaluation procedures, or other relevant details. Finally, it would execute the code and determine whether the reported results can be reproduced, while recording any modifications required to make the experiments run successfully.

Exact reproduction may be infeasible when experiments require substantial computational resources. In such cases, a verifier can instead run appropriately scaled-down experiments designed to test the same underlying claims. Claim-level outcomes can then be categorized as **passed** when the reported result is reproduced, **partially passed** when reproduction requires modifications or is only established through a scaled-down experiment, and **failed** when the reported evidence cannot be reproduced. These outcomes can be aggregated into a paper-level score, with claims weighted according to their importance so that failures on central empirical claims are not obscured by successful reproduction of minor or auxiliary results.

In Paperena, we instantiate this process using the open-source *veritas* verification agent [56]. *veritas* covers most of the stages above, including inspection of the manuscript and accompanying code, execution of experiments, and assessment of whether reported results can be recovered. However, its internal decomposition does not map exactly onto the idealized pipeline: responsibilities are distributed across multiple sub-agents, and it does not perform a fully separate manuscript-only consistency check before consulting the code. We therefore regard *veritas* as a practical approximation to the desired verifier rather than a complete realization of it.

A further limitation is that our current verification procedure focuses primarily on the final manuscript and accompanying implementation. Recent work suggests that research logs and other intermediate workflow artifacts produced by AI scientist systems can expose methodological failures that are difficult to detect from the final paper alone [58]. A stronger verifier could therefore incorporate these artifacts as an additional source of evidence when they are available.

D Novelty Score Verification

This appendix gives the full detail behind the novelty scoring results summarized in Section 4.

D.1 Related work

Most existing approaches treat novelty as gradable rather than binary. One cluster operationalizes it via embedding- or retrieval-based distance to prior literature: "Not Novel Enough" (NNE) [1] models reviewer behaviour through a retrieve-and-compare pipeline validated against real reviewer judgments; the Idea Novelty Checker [80] retrieves, filters, and facet-reranks prior work before judging; and RAG-Novelty [55] reframes the judgment as pairwise rather than absolute. A second cluster scores a paper without external comparison: HSPIM [90] decomposes it section by section, using LLM question-answering to produce per-section novelty and confidence scores aggregated into one paper-level score. A related line applies a similar graded signal earlier in the pipeline, to drive generation rather than evaluate a finished paper: AutoDiscovery [2] uses Bayesian surprise to pick which hypothesis to test next, and Alien Space of Science [9] ranks combinations of idea atoms to maximize coherence while minimizing community availability. The role shift plausibly matters: guiding exploration among many candidates is lower-stakes than serving as a final judgment a generating agent could directly optimize against and exploit.

Beyond these, a wider set of works studies novelty assessment from different angles. NovBench [100] and Sinhaajari et al. [85] benchmark whether LLM novelty judgments agree with human reviewers, the latter reporting a "novelty mirage" where LLM judges over-rate model-generated content specifically. NoveltyBench [107] measures whether a model's outputs are diverse across repeated generations rather than collapsing to similar responses. Padmakumar et al. [70] measures originality relative to a model's training data. Si et al. [83] run a blind comparison of over 100 expert NLP researchers against an LLM ideation agent, finding LLM ideas judged more novel but weaker on feasibility, and flagging failures in LLM self-evaluation and low generation diversity. Gupta and Pruthi [36] show automated plagiarism detectors miss most paraphrased borrowing in AI-generated research, motivating integrity as a check independent from novelty. The work closest to our own setting evaluates AI-generated papers directly: Ravideshik and Kejriwal [76] use three LLM reviewers to score AI-generated papers on originality, treating their mutual agreement as validation rather than comparison to human judgment, with no per-contribution decomposition or retrieval-grounded evidence.

Across all of these, no existing system produces a graded, evidence-grounded, per-section novelty signal validated against real reviewer judgment and none has been tested on both human- and AI-authored work to check whether the same system even transfers between them.

D.2 Scoring layer

The "Not Novel Enough" workflow (NNE) [1]³ extracts a paper's structured claims, retrieves and synthesizes related work, and produces an adversarial, evidence-grounded delta analysis comparing each claimed contribution against retrieved prior work, validated against 182 real ICLR 2025 submissions (86.5% reasoning alignment, 75.3% conclusion agreement with human reviewers). Given NNE's delta-analysis output for a paper, an LLM call reads that text and extracts each claimed contribution's label, one of five: *genuine* (no comparable prior work found), *narrower-than-claimed*,

³<https://ukplab.github.io/eacl2026-assessing-paper-novelty/>

overstated, *already-achieved*, or *speculative*, using the vocabulary NNE’s own delta-analysis already uses, together with a quote drawn directly from that text. Figure 3 shows the full pipeline this sits at the end of.

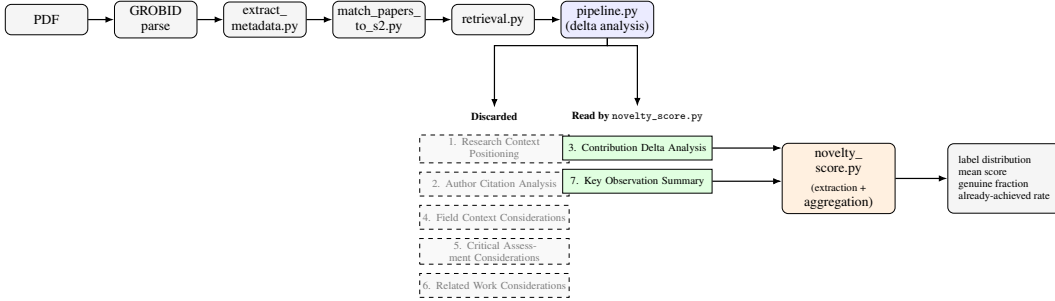


Figure 3: The novelty scoring pipeline. `novelty_score.py`’s extraction step currently reads two of the seven delta-analysis sections (green, right column), *Contribution Delta Analysis* and *Key Observation Summary*.

We report the full label distribution per paper and a mean score (5 for genuine, descending by rank), plus the fraction of contributions labeled genuine. *Already-achieved* is an explicit “this was already done” flag, distinct from the softer *narrower-than-claimed* or *overstated* labels. We also report a ceiling metric: whether any single contribution in the paper is labeled genuine.

D.3 Evaluation results

We first check our scoring layer against NNE’s own released human-annotated dataset [1], which includes binary novel/not-novel verdicts from real reviewers, at two different scales.

Take one pair from this dataset: two papers score almost identically on mean score (3.67 in both cases) despite opposite human verdicts, and the reason is retrieval, not judgment. One reviewer described the novel paper as making a “fresh contribution” that “sets a new standard,” while another recommended rejecting the not-novel paper because its core idea was “already well known” in prior work the reviewer named specifically. The automated delta-analysis cannot make the same distinction: the specific prior work that reviewer cited never appeared in the not-novel paper’s retrieved research landscape, so the system had no evidence the idea was already known, and described its contribution in the same hedged, incremental language it used for the genuinely novel paper. The two are separated only by the ceiling label and genuine count: one genuine contribution for the novel paper, zero for the not-novel one.

At the level of the full dataset, this individual-pair noise coexists with a real aggregate signal. We applied our scoring layer to all 77 papers with a unanimous human verdict (16 novel, 61 not_novel), scoring NNE’s own precomputed delta-analysis text directly. Table 7 reports the result: novel papers score higher than not-novel papers on every metric in the table.

Group	n	mean(mean score)	mean(genuine fraction)	genuine ceiling rate
novel	16	3.678	0.148	56% (9/16)
not_novel	61	3.491	0.078	26% (16/61)

Table 7: Scoring layer applied to all unanimously-classed papers in NNE’s released dataset. Novel papers score higher on every metric in the table. Given the sample size, we treat this as a descriptive, directionally consistent signal rather than a formally significant one.

D.4 Real papers, verbatim copies, and AI Scientist-v2 papers

Having shown that the scoring layer separates novel from not-novel work in aggregate, we now turn to a smaller test where we run retrieval ourselves, on papers of our own choosing: three real NeurIPS calibration papers [50, 69, 51], a verbatim copy of each (title and authors replaced, all other content byte-identical), and the ten AI-Scientist-generated papers. We chose the real papers for the same subject area, predictive calibration and uncertainty quantification, as the AI-generated ones.

This initial test skips full OCR and figure extraction, partly for resource reasons and partly because retrieval is the more pressing limiting factor at this stage. Table 8 reports the group-level results; Table 9 gives the full per-paper breakdown behind them.

Group	n	Mean score	Genuine fraction	Ceiling = genuine
Real (calibration papers)	3	3.64	0.083	1/3
Plagiarized (verbatim copies)	3	3.25	0.083	1/3
AI Scientist-v2	10	3.90	0.284	10/10

Table 8: Novelty scoring layer applied across three groups: three real NeurIPS calibration papers, verbatim copies of each, and the ten AI Scientist-v2 papers. Under the ceiling metric (single best-scoring contribution), all ten AI Scientist-v2 papers and one of three real papers are labeled “genuine.” Genuine fraction and mean score are both higher for the AI Scientist-v2 group than for the real papers.

The Dirichlet-calibration pair [50] shows retrieval working as intended: because the copy is byte-identical to the real paper, both decompose into the same four claimed contributions. Three of them flip from *narrower-than-claimed* (real) to *already-achieved* (copy) once the retrieval stage correctly surfaces the real paper as the copy’s own most-similar prior work, ranked #1 of 91 retrieved candidates; the fourth stays *overstated* in both.

The verified-uncertainty-calibration pair [51] shows what happens when retrieval fails to catch a copy: retrieval never surfaced the real paper as the copy’s closest match (absent from 73 retrieved candidates), so the plagiarism went undetected, and the copy kept a favorable label instead of being downgraded to *already-achieved*, the way retrieval correctly caught in the Dirichlet pair. The result is the wrong direction for a plagiarized copy to land: it scores higher than the real paper (4.00 vs. 3.50).

The trust-under-shift pair sits between the two: for the copy, retrieval never surfaced the real paper as its closest match (absent from all 93 retrieved candidates), the same failure as the verified-uncertainty pair. Unlike that pair, though, the copy still lands on *narrower-than-claimed* rather than *genuine*: the paper in question, Ovadia et al.’s widely-cited study on predictive uncertainty under dataset shift [69], is well-known enough that the model plausibly recognizes it independent of retrieval. That would be a third, distinct mechanism: not retrieval succeeding, and not retrieval failing cleanly, but retrieval failing while the model’s own training-time familiarity with well-known work quietly covers for it. If that’s what’s happening, it’s good news for this specific case, but bad news as a general property: it would mean the system’s apparent reliability is inflated for well-known, highly-cited prior work, exactly the papers retrieval is least likely to need help with, and offers no such backstop for the genuinely obscure or recent prior work that a real novelty check most needs to catch.

Paper	Group	Genuine	Narrower	Already-ach.	Overstated	Speculative	Mean score
Dirichlet calibration	Real	0	3	0	1	0	3.75
Dirichlet calibration	Plagiarized	0	0	3	1	0	2.25
Trust under shift	Real	0	2	0	1	0	3.67
Trust under shift	Plagiarized	0	1	0	1	0	3.50
Verified calibration	Real	1	1	1	1	0	3.50
Verified calibration	Plagiarized	1	2	0	1	0	4.00
P1	AI-generated	1	2	0	2	0	3.80
P2	AI-generated	1	2	1	1	0	3.60
P3	AI-generated	2	3	0	2	0	4.00
P4	AI-generated	2	2	0	1	0	4.20
P5	AI-generated	2	2	0	1	0	4.20
P6	AI-generated	1	1	1	2	0	3.40
P7	AI-generated	2	2	0	2	0	4.00
P8	AI-generated	1	3	0	0	0	4.25
P9	AI-generated	1	3	0	1	1	3.50
P10	AI-generated	2	1	0	2	0	4.00

Table 9: Full per-paper novelty scoring results: real vs. verbatim-copy pairs and the ten AI-Scientist-generated papers. All ten AI-generated papers are labeled “genuine” under the ceiling metric. Paper identifiers follow Table 3.

All ten AI Scientist-v2 papers hit “genuine” at least once, while only one of the three real, peer-reviewed papers does. Mean score and genuine fraction both rank the AI-generated group above the real papers (Table 8), an encouraging but preliminary signal. We see two plausible, non-exclusive explanations, and we cannot distinguish between them here. The first is genuine: AI-generated papers may be reliably good at recombination, systematically finding an unclaimed corner of the literature to build a defensible claim in, even without the abductive capacity for a genuinely surprising leap [105]. The second is an artifact of the pipeline itself: as the pairs above show, a missing retrieval result is never treated as an unresolved unknown, it’s treated as an absence of evidence against the claim, inflating a contribution’s label, whether that means landing on *genuine* outright or just scoring better than accurate retrieval would have produced. The AI-generated papers cite only a handful of sources each, a consequence of how they were generated rather than their subject matter. That gives retrieval a much thinner base to search against, making a missing result more likely, and a missing result to *genuine* regardless of whether the paper is actually novel. This is a limitation of the present test that future work should resolve: testing on papers with denser citation lists, and better retrieval coverage in general, would let us separate a genuine recombination signal from a retrieval artifact.

D.5 Toward a framework

Running an existing system on AI-generated papers is a starting point, not an endpoint; alongside it, we have been sketching what such a framework might look like, bringing together graded distance to prior literature with a sense of how surprising that distance is, rather than collapsing both into a single label.

First, for this initial test we kept the scoring layer simple by design: discrete labels like *genuine* and *narrower-than-claimed* cannot express degree, so a contribution that is 90% distinct from prior work and one that is 60% distinct score identically, and averaging across a paper’s contributions compounds this. Extending the scoring layer with that granularity, for instance via an additional LLM call over the same delta-analysis text, is a natural next step.

Second, whatever label a contribution receives is only as reliable as the retrieval stage that fed it, as the three pairs above show: a found source correctly downgrades a contribution, a missed one does not, and the two failure modes are not equivalent, since a well-known paper’s training-time memory could quietly cover for a missed retrieval while an obscure or recent one has no such backstop. A framework should track retrieval confidence explicitly rather than treating a single retrieval pass as ground truth: a missing result from a thorough, high-coverage search is different from one where retrieval simply couldn’t cover the ground, and only the latter should be flagged as unknown rather than counted as evidence of novelty. We are exploring an improved retrieval pipeline to address this.

Third, none of these metrics know which contribution matters most: a paper’s central claim and a minor side point count equally, so a genuine label on an incidental contribution reads the same as one on the paper’s main thesis. A framework should weight contributions by their importance to the paper’s core claims, not just aggregate them uniformly.

These gaps motivate the graded framework we are aiming to develop. The core idea is a single graded scale of novelty in place of a binary verdict, with the lowest score reserved for plagiarism or fabrication, regardless of how the contribution otherwise reads. Everything else is graded by how novel the difference is, from trivial recombination, to a surprising one, up to a genuine paradigm shift. Orthogonal to this, we also plan to examine contribution type (methodological, empirical, problem-or-paradigm-framing, or theoretical), since a single scale built around comparison to prior work may not fit every type equally well.

E Interestingness Estimation

We use SciJudge-30B-2605 [92] to estimate the interestingness of the 10 papers (under the model’s learned “tastes” from the training community). We consider the win rate (in percentage) of each paper in pairwise round-robin tournaments as a relative estimate of interestingness. In a tournament, each paper is matched against the other 9 papers, with swappings of their order in the prompt to mitigate position biases, totalling 90 matches. Each match is prompted to be judged by SciJudge-30B-2605 with the post-trained prompt template. While the model is fine-tuned with inputs being the publication date, title, and abstract, we consider both abstract-only and full-paper use cases; each forms a separate tournament. The publication date in the input is substituted with the generation date of the paper. We

also report the perplexity of the abstracts and full papers as an absolute surprisal-based estimate of interestingness (Table 4).

From the results, there are no clear correlations (or rank correlations) between perplexity and the judged win rates. A high-perplexity paper can still obtain a 0.0% win rate when faring against other papers. The win rates also differ between the abstract-only tournament and the full-paper tournament.

While quantifying interestingness does not inherently require grounding in validity, as it focuses on attention-attracting effects, some outcomes suggest a risk for undesirable behaviors if these measures are to be used as rewards without careful regularizations. For instance, the title of the most-winning full paper **P9**, also among the most perplexing papers, starts with “Low Regret Is Not Calibration:” This immediately invites readers as it goes against established results on the equivalence between low internal and swap regrets and calibration in the literature. It turns out the paper is on external regrets, and does not upend any established results, which makes the title mildly misleading.

F Generic AI review

F.1 Setup

auto-reviewer⁴ is an open-source reviewing agent that makes eleven sequential model calls per paper: a structured-parsing pass followed by ten review stages—overall understanding, section-level analysis, claim extraction and evidence mapping, novelty, significance and impact, rigor, review planning, drafting, self-critique, and finalization. Each stage receives the PDF as a cached document block together with the outputs of the stages it depends on. The final stage produces a scorecard for soundness, presentation, and contribution (each 1–4), an overall score (1–10), reviewer confidence (1–5), and a seven-point recommendation from *strong reject* to *strong accept*.

We run auto-reviewer with Claude Opus 4.6 at effort=high and web search enabled. Its August 2025 training cutoff predates the ICLR 2026 submission deadline, reducing the risk that the model had seen the calibration submissions or their outcomes. We also exclude openreview.net from the novelty search because OpenReview pages contain the decisions being predicted. The search retains arXiv, venue proceedings, and Papers With Code, and retrieves 4–8 related works per generated paper. All 36 reviews use the same configuration, differing only in the input PDF; all completed without truncation or failure.

Reviewing the ten generated papers cost \$20.81 (\$2.08 per paper; median 15.0 minutes), while the 26 calibration submissions cost \$124.40 (\$4.78 per paper; median 20.4 minutes), largely reflecting differences in paper length.

F.2 Calibration corpus

To interpret the reviewer’s scores, we compare them with scores assigned to real submissions with known outcomes. The reference pool consists of 100 randomly sampled ICLR 2026 submissions, preserving the venue’s oral/accepted/rejected proportions: 2 oral, 37 accepted, and 61 rejected. Author and affiliation information was removed from all PDFs before review.

Under a fixed compute budget, we reviewed 26 papers: 2 oral, 9 accepted, and 15 rejected. The queue was ordered so that each prefix approximately preserved the label distribution of the full pool. We placed both oral papers near the front because, at 2% of the pool, they would otherwise likely fall outside the 26-paper sample. Oral papers are therefore over-represented (7.7% versus 2%). We count oral papers as accepted throughout, giving 11 accepted papers in the main text. These choices increase the accept rate from 39% in the pool to 42.3% in the reviewed sample.

F.3 Results on the generated papers

Table 11 compares these scores with the calibration corpus. The scores are coarse: across all 36 papers, the overall score takes only four distinct values and contribution only two. We therefore report the full distributions rather than significance tests.

Three patterns stand out. First, no generated paper reaches 6/10 overall, whereas 8 of the 11 accepted papers do. The generated papers score either 5 (eight papers) or 4 (two), overlapping with the rejected submissions. Second, contribution separates generated from accepted papers, although the score takes only two values across the full sample. Third, soundness shows the opposite pattern: four generated

⁴<https://github.com/Agents4Academia-AI/auto-reviewer>

ID	Type	Overall (1–10)	Sound. (1–4)	Pres. (1–4)	Contrib. (1–4)	Signif. (1–10)	Recommendation
P1	empirical	5	2	3	2	4	weak reject
P2	theory	5	4	3	2	4	weak accept
P3	empirical	4	2	3	2	4	weak reject
P4	theory	5	4	3	2	4	weak accept
P5	empirical	5	3	3	2	5	weak accept
P6	theory	5	3	4	2	4	weak reject
P7	empirical	4	2	3	2	4	borderline
P8	theory	5	4	3	2	4	borderline
P9	empirical	5	2	3	2	4	weak accept
P10	theory	5	4	3	2	5	weak accept

Table 10: Per-paper scores for the ten generated papers. Significance is the score from the significance and impact stage. All papers score 2/4 on contribution and 4/5 on confidence (omitted).

Dimension	Group	1	2	3	4
Soundness	AI-generated (10)	0	4	2	4
	Real accepted (11)	0	3	8	0
	Real rejected (15)	1	10	4	0
Contribution	AI-generated (10)	0	10	0	0
	Real accepted (11)	0	0	11	0
	Real rejected (15)	0	2	13	0
Presentation	AI-generated (10)	0	0	9	1
	Real accepted (11)	0	0	11	0
	Real rejected (15)	0	1	14	0

Table 11: Distribution of scorecard values across the three groups. All generated papers score 2/4 on contribution, while all accepted papers score 3/4. Soundness shows the opposite pattern at the top of the scale, while presentation varies little across groups.

papers receive 4/4, a score no human-written paper receives. This difference is associated with paper type: the five theory papers score 3, 4, 4, 4, 4, compared with 2, 2, 2, 2, 3 for the five empirical papers.

For the five empirical papers, the reviewer’s reproducibility score from the rigor stage does not align with the replication scores measured by *veritas* [56]. P9 receives the lowest reproducibility score (3/10) despite the highest measured replication (83.3%), while P3 receives 7/10 with 50.0% replication. With only five papers, this comparison is descriptive, but it illustrates that a generic reviewer may not capture properties measured directly by targeted verification.

F.4 Agreement with venue decisions

We assess the reviewer’s recommendations against the known outcomes of the calibration papers. Mapping *weak accept* or higher to accept, the reviewer agrees with the ICLR decision on 61.5% of the 26 submissions.

One calibration paper received *borderline*; it was rejected by ICLR and is mapped to reject. Mapping *borderline* to accept instead reduces accuracy to 57.7% and macro F1 to 0.561. The final stage changed the recommendation produced during review planning for 4 of 26 papers, including one change that crossed the accept/reject boundary. Neither oral paper was distinguished from the other accepted papers: both scored 6/10 overall, 3/4 on soundness, and 3/4 on contribution, and both received *weak accept*, as did eight of the nine non-oral accepted papers.

The reviewer also uses a narrow portion of its available scales. Across all 36 papers, overall scores take values {3, 4, 5, 6}, contribution scores {2, 3}, and confidence is 4/5 for every paper. Of the 36

Metric	Recommendation	Overall score ≥ 6
Accuracy	61.5%	73.1%
Balanced accuracy	65.5%	73.0%
Macro F1	0.606	0.727
F1 (accept)	0.667	0.696
F1 (reject)	0.545	0.759
Precision / recall (accept)	0.526 / 0.909	0.667 / 0.727
Predicted accept rate	73.1%	46.2%

Table 12: Agreement with ICLR decisions on the 26 calibration submissions (actual accept rate 42.3%). The recommendation accepts 10 of 11 accepted papers but also 9 of 15 rejected papers. Thresholding the overall score at 6—the conventional boundary between “marginally above” and “marginally below” acceptance—yields better calibration and higher agreement.

recommendations, 24 are *weak accept* and 8 *weak reject*; no paper received a recommendation above *weak accept*, and only one fell below *weak reject*.

For context, the NeurIPS 2021 consistency experiment, in which 10% of submissions were independently reviewed by two committees, reported 73.4% agreement ($\kappa = 0.314$, $n = 764$ after excluding withdrawals and merging oral, spotlight, and poster into accept). This is not directly comparable to our 61.5%: the NeurIPS experiment measures agreement between two human committees, whereas we compare the reviewer with a single realized venue decision. The accept rates also differ (27.0% versus 42.3%).

F.5 Limitations

The generated set contains 10 papers from a single generator across five topics, so the observed soundness difference between theory and empirical papers rests on five papers of each type. The calibration run covers 26 of the 100 available submissions. Given these sample sizes and the coarse score distributions, we treat the comparisons as descriptive rather than applying significance tests. Each paper was reviewed once, so we also do not measure run-to-run variation.

The calibration corpus has two additional limitations. Although author and affiliation information was removed, rejected submissions retain the anonymous ICLR submission format, including line numbers, whereas accepted papers are camera-ready. This format difference could reveal the outcome. The parsing stage identifies the anonymous submissions as double-blind, but the reviewer’s 61.5% agreement with venue decisions suggests that it does not simply use this cue to predict the label. Finally, ICLR decisions are themselves noisy labels, as illustrated by the NeurIPS consistency experiment; disagreement with the venue therefore does not necessarily indicate reviewer error.